

Energy, Work and Power

History

The concept of **energy** evolved slowly through the 19th century as physicists realised that heat, motion, electricity, and chemical reactions were all different forms of the same thing.

James Prescott Joule (1818–1889), an English brewer and physicist, conducted meticulous experiments showing that mechanical work could be converted into heat. His famous experiment involved a falling weight turning a paddle wheel in water — the friction heated the water. This established the **mechanical equivalent of heat**.

Key figures:

- **Galileo & Newton** — understood kinetic energy ()
- **Thomas Young (1807)** — first used the word "energy" in the modern scientific sense
- **Joule (1840s)** — established the relationship between work and heat
- **Hermann von Helmholtz (1847)** — formalised the **Law of Conservation of Energy**

Myth vs Fact: Perpetual motion machines (devices that produce more energy than they consume) are impossible. The First Law of Thermodynamics (Conservation of Energy) means energy cannot be created or destroyed — only transformed. Every proposed perpetual motion machine either stops or violates this law.

$$\frac{1}{2}mv^2$$

Nature & Applications

-  **Batteries:** Convert chemical energy → electrical energy
-  **Solar panels:** Convert light energy → electrical energy
-  **Generators:** Convert kinetic energy → electrical energy
-  **Food:** Chemical energy in food → thermal energy (warmth) + kinetic energy (movement)
-  **Electric cars:** Battery chemical energy → electrical → kinetic (motion)
-  **Hydroelectric dams:** Gravitational potential energy → kinetic → electrical
-  **Wind turbines:** Kinetic energy of wind → kinetic energy of blades → electrical energy

Theory

Energy

Energy is the capacity to do work. SI unit: **joule (J)**.

Principle of Conservation of Energy:

Energy cannot be created or destroyed. It can only be transferred from one form to another.

Forms of Energy

Form	Symbol	Description	Example
Kinetic		Energy of motion	Moving car, wind
Gravitational potential	or	Energy due to height	Raised weight, water behind a dam
Elastic potential		Energy stored in stretched/compressed material	Drawn bow, compressed spring
Chemical		Energy stored in chemical bonds	Food, fuel, battery
Thermal	or	Energy due to temperature	Hot objects
Electrical		Energy of moving charges	Current in a wire
Nuclear		Energy stored in the nucleus	Fission, fusion
Sound		Energy of vibrating air	Music, speech
Light		Radiant energy	Sun, light bulbs

Kinetic Energy

Where:

- = kinetic energy (J)
- = mass (kg)
- = velocity (m/s)

Gravitational Potential Energy

Where:

- = gravitational potential energy (J)
- = mass (kg)
- = gravitational field strength (N/kg)
- = height above a reference point (m)

Work

Work is done when a force causes an object to move in the direction of the force.

Where:

- = work done (J)
- = force (N)
- = distance moved in the direction of the force (m)

Key idea: of work is done when a force of moves an object .

Power

Power is the rate at which work is done (or energy is transferred).

Where:

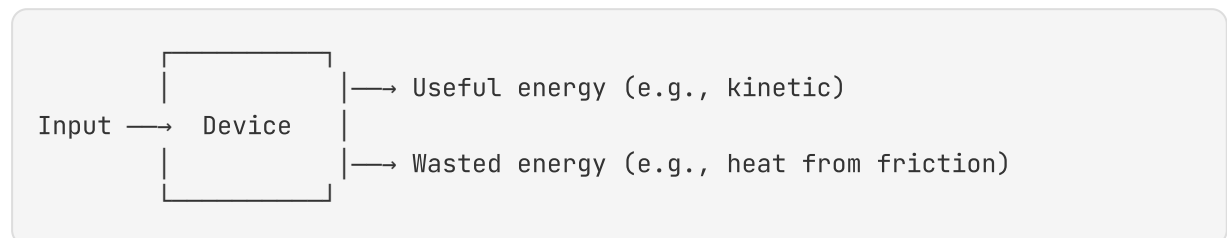
- = power (W)
- = work done (J)
- = energy transferred (J)
- = time taken (s)

Alternative formula for power (using force and velocity):

Where v = constant velocity (m/s) — useful for vehicles climbing hills or overcoming drag.

Energy Transfers (Sankey Diagrams)

A **Sankey diagram** shows the flow of energy through a system — the width of each arrow is proportional to the amount of energy.



Efficiency:

Efficiency can also be calculated using power:

$$E_{\text{total before}} = E_{\text{total after}}$$

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

$$W = Fd$$

$$1 \text{ J} = 1 \text{ Nm}$$

$$P = \frac{W}{t} = \frac{E}{t}$$

$$1 \text{ W} = 1 \text{ J/s}$$

$$P = Fv$$

$$\text{Efficiency} = \frac{\text{useful output energy}}{\text{total input energy}} \times 100\%$$

$$\text{Efficiency} = \frac{\text{useful power output}}{\text{total power input}} \times 100\%$$

E_k E_p E_g E_e E_c Q E_{th} E_e E_n E_s E_l E_k m v E_p m g h W F d 1 J 1 N 1 m P W E t v

Worked Examples

Example 1: Kinetic Energy

A car of mass 1200 kg travels at 20 m/s. What is its kinetic energy?

Solution:

Example 2: Gravitational Potential Energy

A 5 kg book is lifted from the floor to a shelf 2 m high. How much GPE does it gain? (N/kg)

Solution:

Example 3: Work Done

A person pushes a box with a force of 50 N over a distance of 6 m. How much work is done?

Solution:

Example 4: Power

A motor does 2400 J of work in 60 seconds. What is its power?

Solution:

Example 5: Efficiency

A light bulb receives 100 J of electrical energy. It produces 10 J of light energy and 90 J of thermal energy.

(a) What is the efficiency?

(b) What is the percentage efficiency?

Solution (a):

Solution (b):

Example 6: Conservation of Energy (Roller Coaster)

A roller coaster car of mass 500 kg is at the top of a 30 m high hill. It rolls down (ignore friction).

(a) What is its GPE at the top?

(b) What is its speed at the bottom?

Solution (a):

Solution (b):

At the bottom, all GPE has been converted to KE:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 1200 \times 20^2$$

$$= 600 \times 400 = 240\,000 \text{ J} = 240 \text{ kJ}$$

$$E_p = mgh = 5 \times 9.8 \times 2 = 98 \text{ J}$$

$$W = Fd = 50 \times 6 = 300 \text{ J}$$

$$P = \frac{W}{t} = \frac{2400}{60} = 40 \text{ W}$$

$$\text{Efficiency} = \frac{10}{100} = 0.1$$

$$\text{Efficiency} = 0.1 \times 100\% = 10\%$$

$$E_p = mgh = 500 \times 9.8 \times 30 = 147\,000 \text{ J}$$

$$\frac{1}{2}mv^2 = mgh$$

$$\frac{1}{2}v^2 = gh$$

$$v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 30}$$

$$v = \sqrt{588} \approx 24.2 \text{ m/s}$$

$$g = 9.8$$